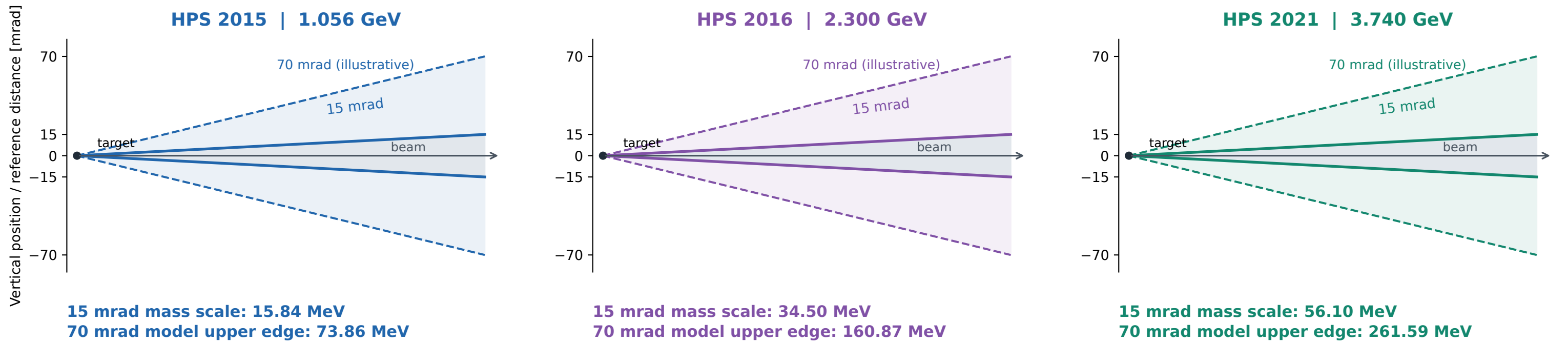
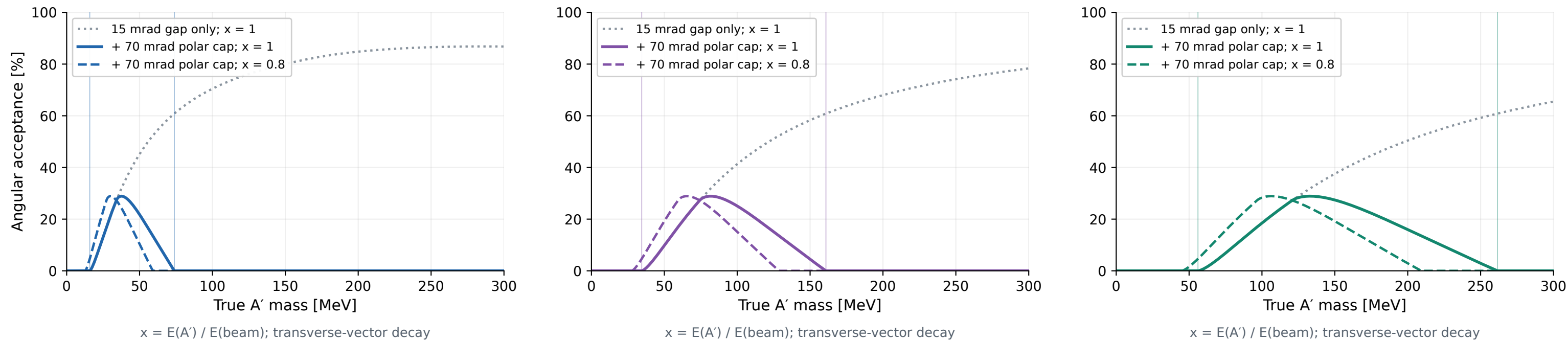


HPS A' angular acceptance and selected mass spectra | v5.7

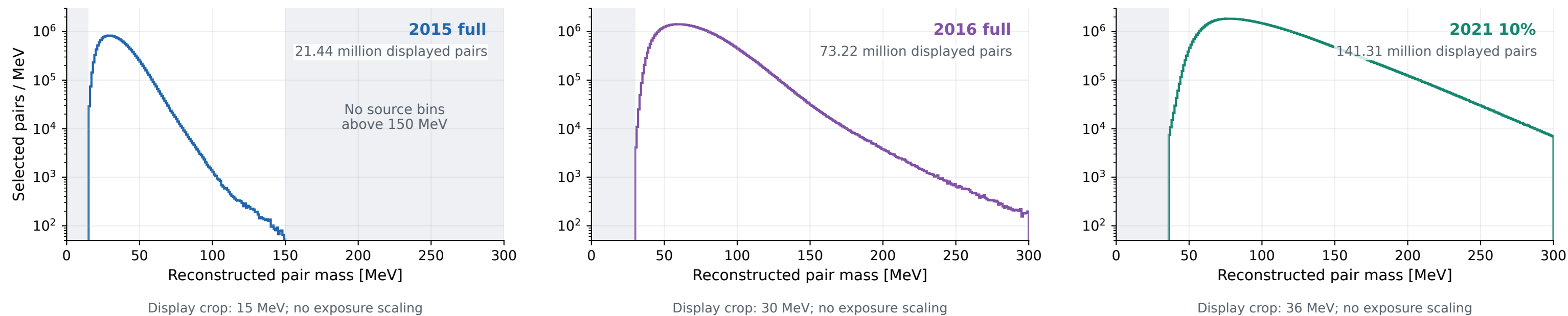
A | Angular scales: equal-energy daughters, prompt on-axis parent, $E(A') = E(\text{beam})$



B | Fraction of A' decays passing an idealized angular window



C | Selected pair counts in the existing data samples



A-B: geometric illustration, not calibrated HPS efficiency. A 15 mrad vertical gap alone gives no upper mass cutoff.
The 70 mrad polar cap is a historical-design-inspired assumption; the actual 2021 tracker and trigger were upgraded.
C: recorded background-dominated spectra, not A' production rates. Display crops and histogram endpoints are not acceptance boundaries.